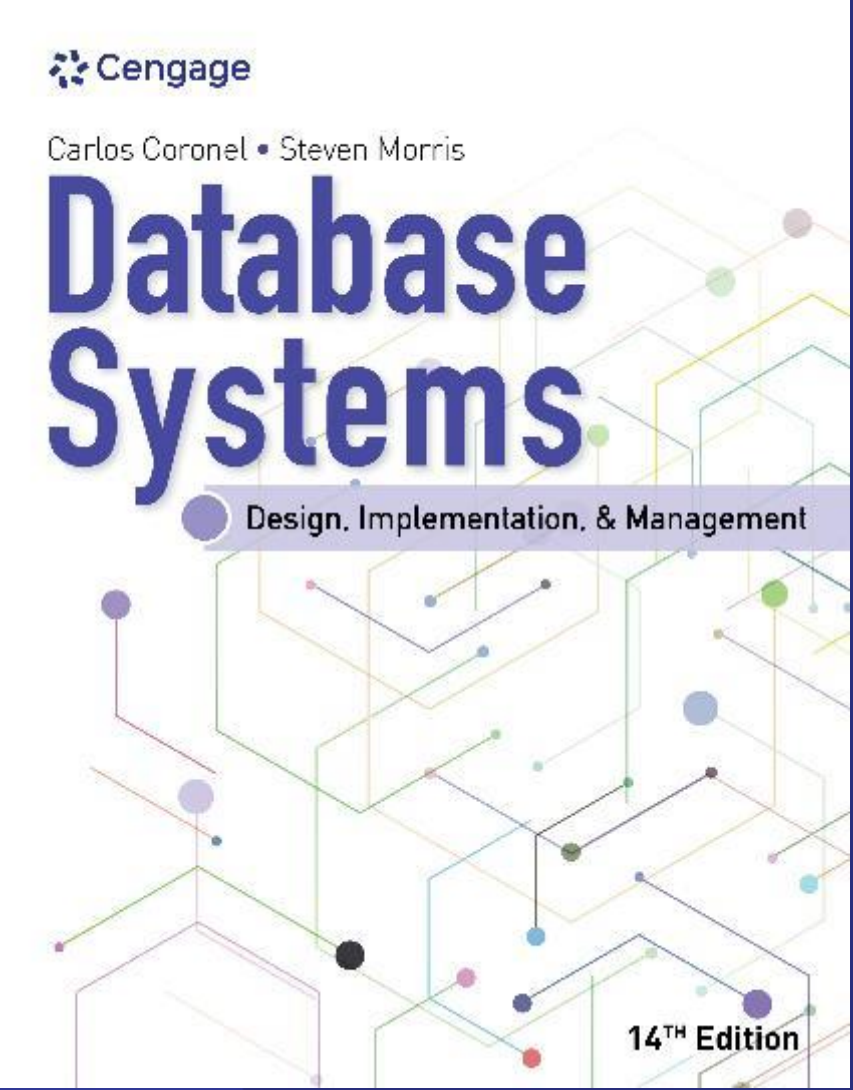
Lecture 11



Database Systems: Design, Implementation, and Management, 14e

Module 6: Normalization of Database Tables

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Chapter Objectives

By the end of this chapter, you should be able to:

1. Explain normalization and its role in the database design process
2. Identify each of the normal forms: 1NF, 2NF, 3NF, BCNF, 4NF, and 5 NF
3. Explain how normal forms can be transformed from lower normal forms to higher normal forms
4. Apply normalization rules to evaluate and correct table structures
5. Identify situations that require denormalization to generate information efficiently
6. Use a data-modeling checklist to check that the ERD meets a set of minimum requirements

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Database Tables and Normalization (1 of 2)

- **Normalization** is a process for evaluating and correcting table structures to minimize data redundancies
  - It reduces the likelihood of data anomalies(insert, delete or update anomalies)
  - Assigns attributes to tables based on determination
- Normalization works through a series of stages called normal forms and the first three are described as follows:
  - First normal form (1NF)
  - Second normal form (2NF)
  - Third normal form (3NF)

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Database Tables and Normalization (2 of 2)

- From a structural point of view, higher normal forms are better than lower normal forms
  - For most purposes in business database design, 3NF is as high as you need to go in the normalization process
- **Denormalization** produces a lower normal form
  - The result of denormalization is increased performance but greater data redundancy

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The Need for Normalization

- Database designers commonly use **normalization** in the following two situations:
  - When designing a new database structure
  - To analyze the relationship among the attributes within each entity and determine if the structure can be improved through normalization
- The main goal of normalization is to eliminate data anomalies by eliminating unnecessary or unwanted data redundancies
- Normalization uses the concept of functional dependencies to identify which attribute determines other attributes

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The Normalization Process (1 of 4)

- The objective of normalization is to ensure that each table conforms to the concept of **well-formed relations** and **has the following characteristics:**
  - Each table represents a single subject
  - Each row/column intersection contains only one value and not a group of values
  - No data item will be unnecessarily stored in more than one table
  - All nonprime attributes in a table are dependent on the primary key
  - Each table has no insertion, update, or deletion anomalies

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The Normalization Process (2 of 4)

- An objective is to ensure that all tables are in at least in 3NF
  - Higher forms, such as 5NF and DKNF are not likely to be encountered in business environment
- The normalization process works one relation at a time, identifying the dependencies of a relation (table)

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The Normalization Process (3 of 4)

Normal Forms

Table 6.2		
Normal Form	**Characteristic**	**Section**
First normal form (1NF)	Table format, no repeating groups, and PK identified	6-3a
Second normal form (2NF)	1NF and no partial dependencies	6-3b
Third normal form (3NF)	2NF and no transitive dependencies	6-3c
Boyce-Codd normal form (BCNF)	3NF and every determinant is a candidate key (special case of 3NF)	6-6a
Fourth normal form (4NF)	BCNF and no independent multivalued dependencies	6-6b
Fifth normal form (5NF or PJNF)	4NF and cannot have lossless decomposition into smaller tables	6-6c

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The Normalization Process (4 of 4)

- A **partial dependency** exists when there is a functional dependence in which the determinant is only part of the primary key
  - The assumption is that there is only one candidate key
  - Partial dependencies tend to be straightforward and easy to identify
- A **transitive dependency** exists when the attribute is dependent on another attribute that is not part of the primary key
  - Transitive dependencies are more difficult to identify among a set of data
  - They occur only when a functional dependence exists among non-prime attributes

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Knowledge Check Activity 6-1

- **What is normalization?**

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Knowledge Check Activity 6-1: Answer

- **What is normalization?**
- **Answer:** Normalization is the process for assigning attributes to entities. Properly executed, the normalization process eliminates uncontrolled data redundancies, thus eliminating the data anomalies and the data integrity problems that are produced by such redundancies.

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Conversion to First Normal Form (1NF) (1 of 3)

- A repeating group derives its name from the fact that a group of multiple entries of the same type can exist for any single key attribute occurrence
  - Normalizing the table structure will reduce data redundancies
- Normalization starts with the following **three-step procedure:**
  - Eliminate the repeating groups
  - Identify the primary key
  - Identify all dependencies

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Conversion to First Normal Form (1NF) (2 of 3)

- A dependency diagram depicts all dependencies found within given table structure
  - It helps to get an overview of all relationships among table's attributes
  - Their use makes it less likely that an important dependency will be overlooked
- The term 1NF describes the tabular format in which the following occur:
  - All key attributes are defined
  - There are no repeating groups in the table
  - All attributes are dependent on the primary key
- All relational tables satisfy 1NF requirements
- Some tables contain partial dependencies

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Conversion to First Normal Form (1NF) (3 of 3)

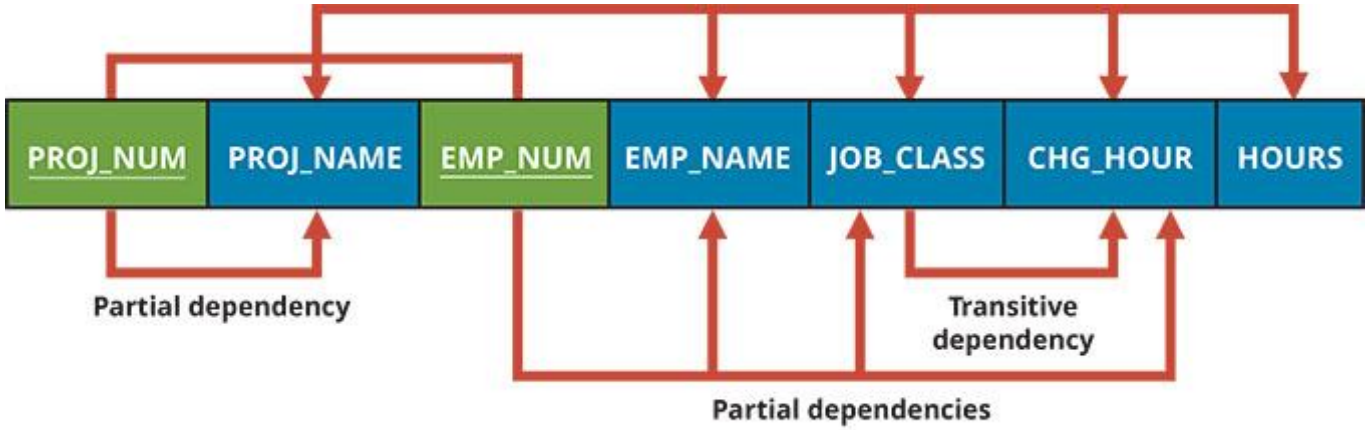


Figure 6.3 First Normal Form (1NF) Dependency Diagram

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Conversion to Second Normal Form (2NF) (1 of 2)

- Conversion to **2NF** occurs only when the 1NF has a composite primary key
  - If the 1NF has a single-attribute primary key, then the table is automatically in 2NF
- The **1NF-to-2NF conversion** is simple, you take the following steps:
  - Step 1: Make new tables to eliminate partial dependencies
  - Step 2: Reassign corresponding dependent attributes
- A table is in **2NF** under the following circumstances :
  - When it is in 1NF
  - When it includes no partial dependencies

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Conversion to Second Normal Form (2NF) (2 of 2)

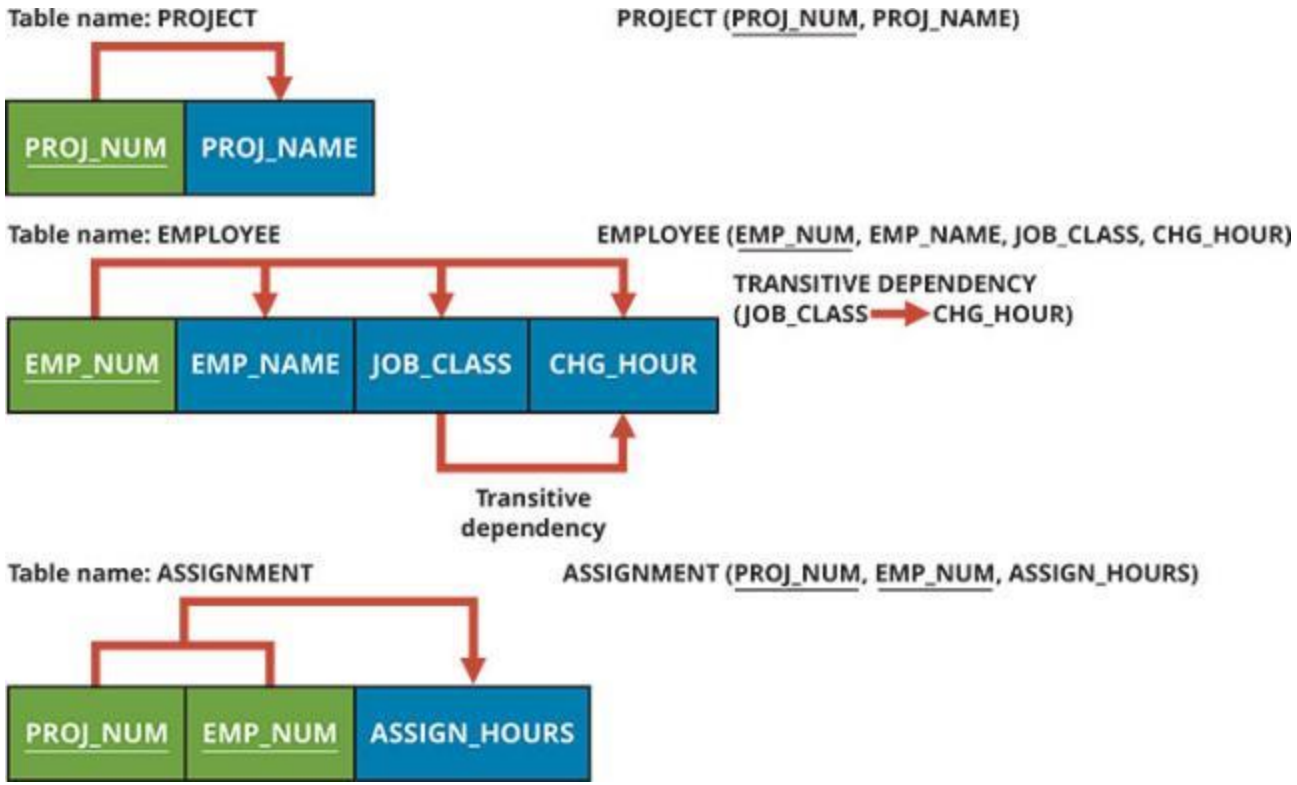


Figure 6.4 Second Normal Form (2NF) Conversion Results

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Conversion to Third Normal Form (3NF) (1 of 2)

- The data anomalies created by the database organization shown in Figure 6.4 are easily eliminated by completing the following two steps:
  - Step 1: Make new tables to eliminate transitive dependencies
  - Step 2: Reassign corresponding dependent attributes
- A table is in **3NF** under the following circumstances:
  - When it is in 2NF
  - When it contains no transitive dependencies

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Conversion to Third Normal Form (3NF) (2 of 2)

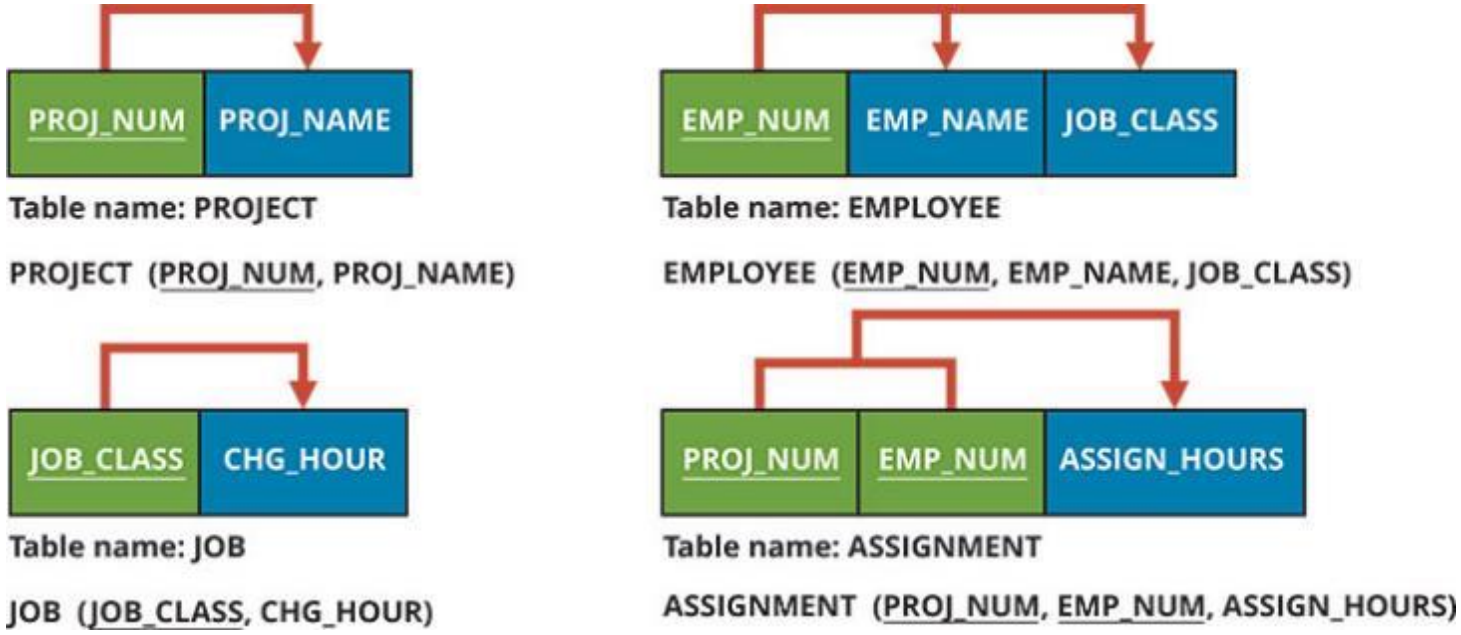


Figure 6.5 Third Normal Form (3NF) Conversion Results

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Improving the Design (1 of 2)

- The following are various types of issues you need to address to produce a **good normalized set of table:**
  - Minimize data entry errors
  - Evaluate naming conventions
  - Refine attribute atomicity
    - An **atomic attribute** is an attribute that cannot be further subdivided
    - **Atomicity** is a characteristic an attribute that cannot be divided into smaller units
  - Identify new attributes
  - Identify new relationships

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Knowledge Check Activity 6-2

- **When is a table in 3NF?**

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Knowledge Check Activity 6-2: Answer

- **When is a table in 3NF?**
- **Answer:** A table is in 3NF when it is in 2NF and it contains no transitive dependencies.

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Surrogate Key Considerations

- **Surrogate keys** are used by designers when the primary key is considered to be unsuitable
- A surrogate key is a system-defined attribute generally created and managed via the DBMS
- Usually it is a numeric value which is automatically incremented for each new row

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Summary

Now that the lesson has ended, you should be able to:

1. Explain normalization and its role in the database design process
2. Identify each of the normal forms: 1NF, 2NF, 3NF, BCNF, 4NF, and 5 NF
3. Explain how normal forms can be transformed from lower normal forms to higher normal forms
4. Apply normalization rules to evaluate and correct table structures
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Link: Article on Normalization

https://www.databasestar.com/database-normalization/

Normalization in SQL (1NF - 5NF): A Beginner's Guide | DataCamp

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Concepts

- Normalization
- Data Anomalies
- 1NF, 2NF and 3NF
- Atomic Attribute, Atomicity
- Partial/ Transitive Dependencies
- Surrogate Keys

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Questions?



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